$$1 + 2N_i(T) = 1 + \frac{2}{\mathrm{e}^{\omega_i/k_\mathrm{B}T} - 1}$$
$$= \frac{\mathrm{e}^{\omega_i/k_\mathrm{B}T} + 1}{\mathrm{e}^{\omega_i/k_\mathrm{B}T} - 1} = \coth\left(\frac{\omega_i}{2k_\mathrm{B}T}\right). \tag{5.14}$$